

Tutorial 8 Gravitational Fields Suggested Solutions

D1 (a) $F = \frac{GMm}{r^2}$

M, m are the masses of 2 point masses

F is the gravitational force of attraction between the 2 point masses

r is the distance between the 2 point masses

G is the gravitational constant

- (b)** Gravitational field strength at a point is defined as the gravitational force per unit mass at that point.

$$g = \frac{F}{m} = \frac{GMm}{r^2} \left(\frac{1}{m} \right) = \frac{GM}{(R+h)^2}$$

where $(R+h)$ is the distance between the centre of the planet to the point that is height h above its surface.

(c) $\frac{R_E}{R_M} = 3.7$ and $\frac{M_E}{M_M} = 81$

$$g_E = \frac{GM_E}{R_E^2} = 9.8 \text{ N kg}^{-1}$$

(i)

$$g_M = \frac{GM_M}{R_M^2}$$

$$\frac{g_M}{g_E} = \frac{GM_M}{R_M^2} \times \frac{R_E^2}{GM_E} = \frac{M_M}{M_E} \times \frac{R_E^2}{R_M^2} = \frac{1}{81} \times (3.7)^2$$

$$g_M = \frac{1}{81} \times (3.7)^2 \times 9.8 = 1.656 = 1.66 \text{ N kg}^{-1}$$

- (ii)** Let d be the distance between this point and the centre of the Earth.
Let D be the distance between the centres of the Earth and the Moon.
Given: $D = 3.84 \times 10^8 \text{ m}$

At the point where the combined gravitational field strength is zero,

$$g_{E_0} = g_{M_0}$$

$$\frac{GM_E}{d^2} = \frac{GM_M}{(D-d)^2}$$

$$\frac{D-d}{d} = \sqrt{\frac{M_M}{M_E}}$$

$$D = d \sqrt{\frac{M_M}{M_E}} + d$$

$$= d \left(\sqrt{\frac{M_M}{M_E}} + 1 \right)$$

$$d = \frac{D}{\left(\sqrt{\frac{M_M}{M_E}} + 1 \right)} = \frac{3.84 \times 10^8}{\sqrt{\frac{1}{81}} + 1} = 3.456 \times 10^8 = 3.46 \times 10^8 \text{ m}$$

D2 (a) (i)
$$F_g = \frac{GMm}{r^2}$$

$$= \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(1.00)}{(6370 \times 10^3)^2}$$

$$= 9.83 \text{ N}$$

(ii) The centripetal force required is

$$F_c = mr\omega^2$$

$$= (1.00)(6370 \times 10^3) \left(\frac{2\pi}{24 \times 60 \times 60} \right)^2$$

$$= 0.0337 \text{ N}$$

(iii) $(F_g - T)$ is the resultant force on the mass and it provides the centripetal force.

$$F_g - T = F_c$$

$$T = 9.83 - 0.0337 = 9.80 \text{ N}$$

(b) (i) $F_g = ma_g$

$$a_g = \frac{9.83}{1.00} = 9.83 \text{ m s}^{-2}$$

(ii) $a_T = \frac{9.80}{1.00} = 9.80 \text{ m s}^{-2}$

- (c)**
- At the Equator, the student undergoes circular motion.
 - The student is measuring the acceleration of free fall, which is the relative acceleration of the falling ball accelerating due to the gravitational force, with respect to his (the student's) acceleration at the Equator, which is his centripetal acceleration.
 - The measured acceleration of free fall will thus be smaller than that due to gravity ($a_{\text{freefall}} = a_{\text{grav}} - a_{\text{centri}}$).
 - Hence, the statement is wrong.

OR

- The Earth's gravitational field strength at the Equator is responsible for the centripetal acceleration of the falling ball and its free fall acceleration i.e. $a_{\text{grav}} = a_{\text{centri}} + a_{\text{freefall}}$.
- The measured acceleration of free fall will thus be smaller than the acceleration due to gravity ($a_{\text{freefall}} = a_{\text{grav}} - a_{\text{centri}}$).
- Hence, the statement is wrong.

D3 (a) (i) Angular velocity is the rate of change of angular displacement of the object as it travels in a circle.

(ii) Period of the Earth round the Sun = 365days $\times$ 24h $\times$ 60mins $\times$ 60s

$$\omega = \frac{2\pi \text{ rad}}{T} = \frac{2\pi \text{ rad}}{(365 \times 24 \times 60 \times 60) \text{ s}} = 1.992 \times 10^{-7} = 1.99 \times 10^{-7} \text{ rad s}^{-1}$$

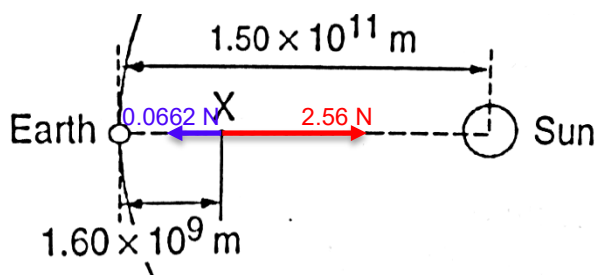
(b) (i) 1.

$$F_E = \frac{GM_E m}{r_E^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(425)}{(1.60 \times 10^9)^2} = 0.06622 = 0.0662 \text{ N}$$

2.

$$F_S = \frac{GM_S m}{r_S^2} = \frac{(6.67 \times 10^{-11})(1.99 \times 10^{30})(425)}{(1.50 \times 10^{11} - 1.60 \times 10^9)^2} = 2.562 = 2.56 \text{ N}$$

(ii)



(iii) 1. resultant force $F_R = 2.562 - 0.06622 = 2.496 = 2.50 \text{ N}$ towards the Sun

2. acceleration $a = \frac{F_R}{m} = \frac{2.496}{425} = 5.873 \times 10^{-3} = 5.87 \times 10^{-3} \text{ m s}^{-2}$

(iv) $a = \omega^2 r$

$$\omega = \sqrt{\frac{a}{r}} = \sqrt{\frac{5.873 \times 10^{-3}}{1.50 \times 10^{11} - 1.60 \times 10^9}} = 1.989 \times 10^{-7} = 1.99 \times 10^{-7} \text{ rad s}^{-1}$$

(v) Both the satellite and the Earth have the same period of 1 year around the Sun, as they both have the same angular velocity as calculated in (a)(ii) and (b)(iv). The satellite and the Earth will always lie along the same straight line with the Sun as they orbit around the Sun.

As the satellite is meant to observe the Sun continuously, orbiting around the Sun will ensure that the satellite will always be able to observe the Sun unobstructed. If the satellite orbits around the Earth, there will be times when the satellite is on the other side of Earth such that the Earth is between the satellite and the Sun. When this happens, the satellite will not be able to observe the Sun fully.

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- (vi)
1. As the satellite is between the Earth and the Sun, the satellite needs to have a very powerful transmitter so that its signal is strong enough to be distinguished from the electromagnetic waves emitted by the Sun.
 2. This orbit is unstable as any disturbance (solar flare, solar radiation pressure, impact with space dust) will cause the satellite to move from this orbit. The satellite will need a large fuel tank to keep it in this orbit.
 3. The satellite is even further from the Earth compared to the moon (average distance of moon from Earth is 3.84×10^8 m). Thus it is expensive and difficult to send a satellite to this orbit because of the long distance.
 4. At least 3 base stations are required at different locations on Earth to receive the signals from the satellite continuously. If there is only one base station, it will not be able to receive any signal from the satellite when the Earth rotates such that the base station is not in the line of sight of the satellite. The continuous observation of the Sun is thus not possible.

Fun facts:

The only satellite that can be repaired in space is the Hubble Space Telescope as it is in low earth orbit and is designed to be serviced by the Space Shuttle. As the Space Shuttle program ended in 2011, there is currently no ability to repair satellites in orbit.

The Deep Space Network is NASA's international array of giant radio antennas that supports interplanetary spacecraft missions, plus a few that orbit Earth.

<https://deepspace.jpl.nasa.gov/about/>

D4 (a) (i) $g_P = \frac{GM}{x^2}$

(ii) $E_P = -\frac{GMm}{x}$

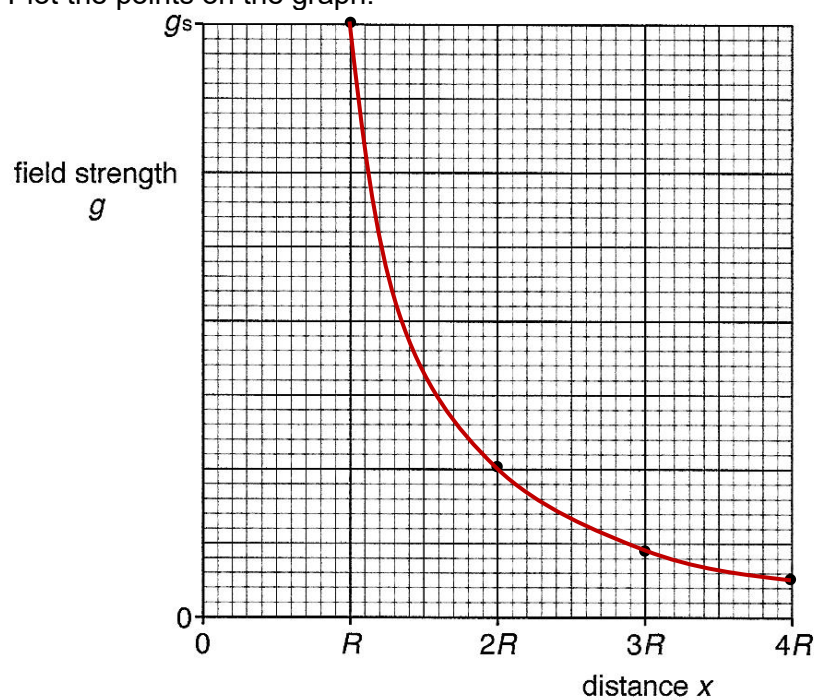
(b) (i) $g_s = \frac{GM}{R^2}$

$$g_{2R} = \frac{GM}{(2R)^2} = \frac{1}{4}g_s$$

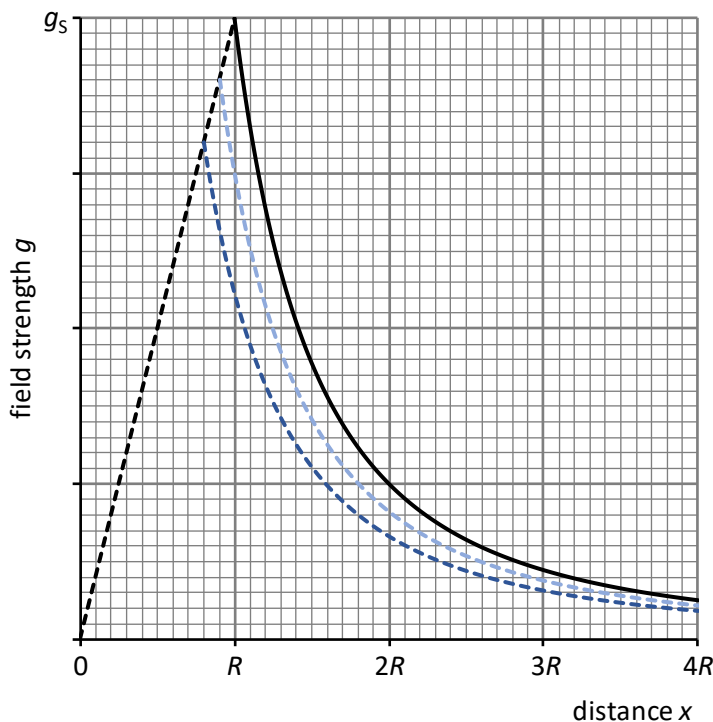
$$g_{3R} = \frac{GM}{(3R)^2} = \frac{1}{9}g_s$$

$$g_{4R} = \frac{GM}{(4R)^2} = \frac{1}{16}g_s$$

Plot the points on the graph.



(ii)



Since the mass is lost uniformly from the surface, the sphere remains spherical but with a smaller radius.

The gravitational field strength on the surface is

$$g'_s = \frac{GM}{r_s^2} = \frac{G \cdot \frac{4}{3}\pi r_s^3 \rho}{r_s^2} = \frac{4}{3}\pi G \rho r_s$$

This shows that g'_s is dependent on the radius r_s of the sphere (straight dotted black line).

So while the mass is gradually lost, the graph would gradually shift towards lower g'_s values, where the starting of the graph is always along the straight dotted black line, with $1/x^2$ variation.

- (c) (i) Star A and star B attract each other by Newton's law of gravitation. By Newton's third law, the force of attraction on A by B is equal in magnitude and opposite in direction to the force of attraction on B by A. The magnitude of this gravitational force of attraction is $\frac{GM_A M_B}{d^2}$.

Since the only force acting on each star is this gravitational force of attraction, the gravitational force is also the centripetal force on each star.

Hence the magnitude of the centripetal force on each star is the same.

(ii)

$$\begin{aligned} \omega &= \frac{2\pi}{T} \\ &= \frac{2\pi}{4.0 \times 365 \times 24 \times 60 \times 60} \\ &= 4.981 \times 10^{-8} = 4.98 \times 10^{-8} \text{ rad s}^{-1} \end{aligned}$$

- (iii) Consider star A and B in circular motion about the same centre of orbit. Star A and B will have the same angular speed ω .

$$\text{gravitational force on A} = \text{centripetal force on A} = M_A \omega^2 r_A$$

$$\text{gravitational force on B} = \text{centripetal force on B} = M_B \omega^2 r_B$$

Since the gravitational force on A = gravitational force on B,

$$M_A \omega^2 r_A = M_B \omega^2 r_B$$

$$\frac{r_A}{r_B} = \frac{M_B}{M_A} = \frac{1}{3.0}$$

Since the ratio of r_A to r_B is 1:3,

$$r_A = \frac{1}{4} d = \frac{1}{4} (3.0 \times 10^{11}) = 7.50 \times 10^{10} \text{ m}$$

Note:

centre of mass of the binary star system about which each star rotates is such that

$$M_A r_A = M_B r_B \text{ which also gives } \frac{r_A}{r_B} = \frac{M_B}{M_A} = \frac{1}{3.0}$$

- (d) Consider star A in circular motion.
The gravitational force on it provides for the centripetal force.

$$\frac{GM_A M_B}{d^2} = M_A \omega^2 r_A$$

$$M_B = \frac{\omega^2 r_A d^2}{G}$$

$$= \frac{(4.981 \times 10^{-8})^2 (7.50 \times 10^{10})(3.0 \times 10^{11})^2}{6.67 \times 10^{-11}}$$

$$= 2.512 \times 10^{29} = 2.51 \times 10^{29} \text{ kg}$$

$$M_A = 3.0 M_B$$

$$= 3.0 (2.512 \times 10^{29})$$

$$= 7.536 \times 10^{29} = 7.54 \times 10^{29} \text{ kg}$$

- (e) Observe the light intensity of the binary star system.

Minimum intensity occurs when the line joining the two stars is along the line of sight from Earth. At these positions, the stars are directly in front or behind each other and thus blocking each other's intensity the most.

The time between two minimum intensities is half a period. Multiply this time by 2 to obtain the period.

D5 (a)
$$g = \frac{GM}{r^2}$$

$$= \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.37 \times 10^6 + 0.12 \times 10^6)^2}$$

$$= 9.47 \text{ N kg}^{-1}$$

(b) When in orbit, both the spacecraft and the astronaut are experiencing the same acceleration and, hence, they are falling at the same rate. Therefore, the astronaut does not experience any contact force from the floor of the spacecraft and he seems to float. This is known as apparent weightlessness.

(c) (i) Gravitational potential at a point in a gravitational field is defined as the work done per unit mass by an external force in bringing a small test mass from infinity to that point.

- (ii)**
- Gravitational potential (or potential energy) at infinity is zero.
 - Since gravitational force is attractive in nature, to bring a mass from infinity to a point in the gravitational field, the direction of the external force is opposite to the direction of displacement of the mass. (We can imagine the external force supporting the mass as it is being lowered towards the Earth.)
 - This results in negative work done per unit mass by the external force.
 - Hence, based on its definition, gravitational potential (or potential energy) has a negative value.

(d)
$$\Delta U = m\Delta\phi_i$$

$$= m(\phi_f - \phi_i)$$

$$= m\left(\left(-\frac{GM}{r_s}\right) - \left(-\frac{GM}{r_E}\right)\right)$$

$$= (6.67 \times 10^{-11})(5.98 \times 10^{24})(3000)\left(-\frac{1}{0.12 \times 10^6 + 6.37 \times 10^6} + \frac{1}{6.37 \times 10^6}\right)$$

$$= (6.67 \times 10^{-11})(5.98 \times 10^{24})(3000)\left(-\frac{1}{6.49 \times 10^6} + \frac{1}{6.37 \times 10^6}\right)$$

$$= 3.47 \times 10^9 \text{ J}$$

D6 The gradient of the tangent to the curve at $r = R_E$ represents the value of the gravitational force on the mass m by the Earth at the surface of the Earth.

(a) Point B.

At the top of its trajectory at distance R , kinetic energy of the rocket is zero. Hence its total energy is equal to its potential energy. Point B is the point at which the potential energy graph and line PQ coincides.

(b) Point C.

If the rocket is already falling back to Earth at distance R (i.e. the top of its trajectory is at a height greater than R), its total energy must be greater than at point B as kinetic energy is now a positive non-zero value. However, it cannot be point D or E as these energies will allow the rocket to reach infinity and not fall back towards the Earth.

(c) Point D.

If the rocket has just enough energy to reach infinity, its total energy is zero.

Point E.

If the rocket has more than 'just enough energy' due to the launch speed being more than the minimum escape velocity, its total energy is positive and is non-zero at infinity.

D7 $U = -\frac{Gm_E m}{r}$

- (a) The gravitational potential at any point between Earth and Moon is given by $\phi = -\frac{GM_M}{r_M} - \frac{GM_E}{r_E}$ where r_M and r_E is the distance from the centre of Moon and Earth respectively.

At a point near to either Moon or Earth, where r_M or r_E is very small, the gravitational potential is very negative. If we move away from Moon towards Earth, r_M increases while r_E decreases. This means that the potential will increase (become less negative) and peak at a point before decreasing towards Earth (becoming more negative again near Earth). The asymmetry is due to the different masses of Moon and Earth.

- (b) The gravitational field strength at any point between the Moon and the Earth can be found from the gradient of the potential-distance graph at that point. Multiplying the value of the gradient by the mass of the tektite gives the resultant gravitational force on the tektite at that point.
- (c) Since the gradient of the tangent at point P is zero, the resultant gravitational force on the tektite when it is at point P is zero.

$$F_M = F_E$$

$$\frac{GM_M m}{x^2} = \frac{GM_E m}{y^2} \quad \text{where } m \text{ is the mass of the tektite}$$

$$\frac{x^2}{y^2} = \frac{M_M}{M_E} = \frac{7.4 \times 10^{22}}{6.0 \times 10^{24}}$$

$$\frac{x}{y} = 0.11$$

- (d) From the surface of the Moon to point P, there is an increase in the gravitational potential of

$$\Delta\phi = \phi_f - \phi_i = (-1.3 \times 10^6) - (-3.9 \times 10^6) = 2.6 \times 10^6 \text{ J kg}^{-1}.$$

To overcome this potential barrier, the tektite must be projected with an amount of kinetic energy that is equivalent to the corresponding increase in gravitational potential energy.

For the tektite to be just able to reach point P from the surface of the Moon (i.e. final kinetic energy of tektite at P is zero),

decrease in K.E. = increase in P.E.

$$\frac{1}{2}mv_o^2 - 0 = m(\phi_f - \phi_i)$$

$$v_o = \sqrt{2(\phi_f - \phi_i)}$$

$$= \sqrt{2(2.6 \times 10^6)}$$

$$= 2.3 \times 10^3 \text{ m s}^{-1}$$

After point P, the gravitational potential decreases all the way to Earth. The tektite will accelerate towards Earth under the action of a stronger gravitational force due to Earth.

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- (e)
- Since atmospheric resistance is neglected, the total energy (K.E. + G.P.E.) of the tektite remains constant.
 - As the tektite moves from the Moon to point P, its K.E. decreases by the same amount as the increase in its G.P.E. As the tektite moves from point P to Earth, G.P.E. is converted back to K.E.
 - Since the change in G.P.E. is much greater when it moves from point P to Earth compared to when it moves from the Moon to point P, the tektite will have a much higher K.E. when it reaches Earth.
 - Hence, the tektite will reach the Earth's surface with a speed greater than the speed of projection from the Moon.

- D8 (a) (i)** The gravitational force on the satellite provides the centripetal force as it orbits the planet.

$$\begin{aligned}\frac{GMm}{r^2} &= m \frac{v^2}{r} \\ v_{\text{orbital}} &= \sqrt{\frac{GM}{r}} \\ &= \sqrt{\frac{(6.67 \times 10^{-11})(5.7 \times 10^{24})}{7.0 \times 10^6}} \\ &= 7.37 \times 10^3 \text{ m s}^{-1}\end{aligned}$$

- (ii)** By conservation of energy,
initial total energy at the surface = final total energy at infinity

$$\begin{aligned}-\frac{GMm}{R} + \frac{1}{2}mv^2 &= 0 \\ \frac{GMm}{R} &= \frac{1}{2}mv^2 \\ v_{\text{escape}} &= \sqrt{\frac{2GM}{R}} \\ &= \sqrt{\frac{2(6.67 \times 10^{-11})(5.7 \times 10^{24})}{6500 \times 10^3}} \\ &= 1.08 \times 10^4 \text{ m s}^{-1}\end{aligned}$$

(b) $M = V\rho = \left(\frac{4}{3}\pi R^3\right)\rho$

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{8\pi G\rho}{3}} \times R$$

$$v_{\text{esc}} \propto R$$

Therefore, v_{esc} will be reduced by a factor of 10^3 .

- (c)** From answers to **(a)** and **(b)**, v_{esc} will be about 11 m s^{-1} for smaller planets. Since the speed of most gas particles would exceed that, they would escape from the surface of the planet easily. (Average speed of a gas particle is 500 m s^{-1} at room temperature.)

- D9 (a) (i)** The gravitational force on the satellite by the Earth provides the centripetal force for the circular motion of the satellite.

$$\frac{GMm}{R^2} = \frac{mv^2}{R} \quad \text{where } m \text{ is the mass of the satellite}$$

$$v^2 = \frac{GM}{R}$$

(ii)

$$E_k = \frac{1}{2}mv^2$$

$$E_k = \frac{1}{2}m\left(\frac{GM}{R}\right)$$

$$E_k = \frac{GMm}{2R}$$

(b) (i)

$$E_p = -\frac{GMm}{R}$$

(ii)

$$E_t = E_k + E_p$$

$$E_t = \frac{GMm}{2R} + \left(-\frac{GMm}{R}\right)$$

$$E_t = -\frac{GMm}{2R}$$

Note that the total energy of a bound system is negative.

- (c) (i)** Since the satellite is losing energy, total energy E_t decreases to become more negative.

- (ii)**
1. From the expression for E_t , E_t decreases when R decreases.
 2. From the expression for E_k , E_k increases when R decreases. Therefore speed increases.

- D10 (a) (i)** The gravitational force acting on the satellite in orbit provides the centripetal force.

$$\frac{GMm}{R^2} = \frac{mv^2}{R}$$

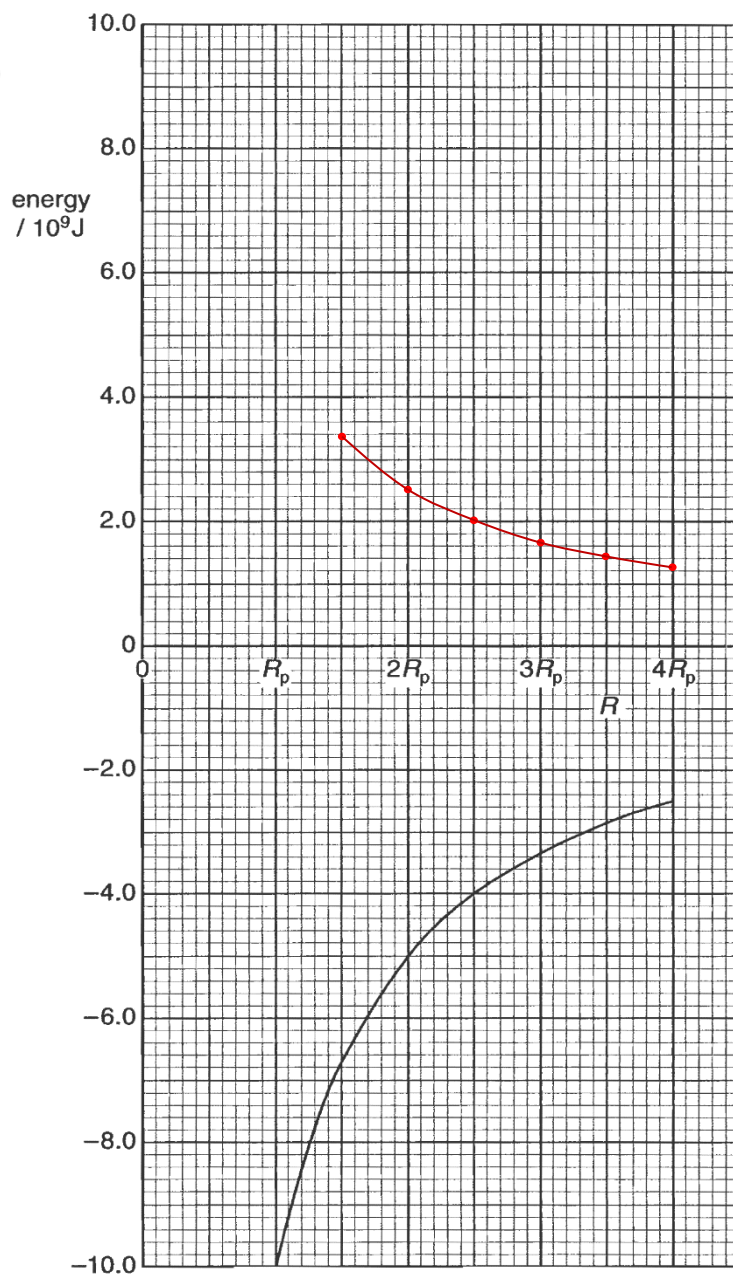
$$mv^2 = \frac{GMm}{R}$$

$$\text{kinetic energy of the satellite, } \frac{1}{2}mv^2 = \frac{1}{2} \frac{GMm}{R}$$

- (ii)** gravitational potential energy of satellite = $-\frac{GMm}{R}$

$$\frac{\text{gravitational potential energy of satellite}}{\text{kinetic energy of satellite}} = \left(-\frac{GMm}{R} \right) / \left(\frac{1}{2} \frac{GMm}{R} \right) = -2$$

- (b) (i)** K.E. of satellite =
 $-\frac{1}{2}(\text{G.P.E. of satellite})$



- (ii) From the graph,
K.E. at $R = 4R_p$ is 1.3×10^9 J

$$\Rightarrow \frac{1}{2}mv_1^2 = 1.3 \times 10^9$$

$$\Rightarrow v_1 = \sqrt{\frac{2(1.3 \times 10^9)}{1600}} = 1.275 \times 10^3 \text{ m s}^{-1}$$

K.E. at $R = 2R_p$ is 2.5×10^9 J

$$\Rightarrow \frac{1}{2}mv_2^2 = 2.5 \times 10^9$$

$$\Rightarrow v_2 = \sqrt{\frac{2(2.5 \times 10^9)}{1600}} = 1.768 \times 10^3 \text{ m s}^{-1}$$

change in orbital speed,

$$v_2 - v_1 = (1.768 \times 10^3) - (1.275 \times 10^3) = 493 \text{ m s}^{-1}$$

C1 For a uniform solid sphere,

$$M = \frac{4}{3}\pi R^3 \rho.$$

Gravitational field strength acting on m inside the sphere, distance r from the centre,

$$\begin{aligned} g &= \frac{G\left(\frac{4}{3}\pi r^3 \rho\right)}{r^2} \\ &= \frac{G\left(\frac{4}{3}\pi r^3\right)\left(\frac{M}{\frac{4}{3}\pi R^3}\right)}{r^2} \\ &= \frac{GMr}{R^3} \end{aligned}$$

$$\begin{aligned} \phi &= \int_{\infty}^R \frac{GM}{r'^2} dr' + \int_R^r \frac{GMr'}{R^3} dr' \\ &= \left[-\frac{GM}{r'} \right]_{\infty}^R + \left[\frac{GMr'^2}{2R^3} \right]_R^r \\ &= -\frac{GM}{R} + \frac{GMr^2}{2R^3} - \frac{GMR^2}{2R^3} \\ &= \frac{GMr^2}{2R^3} - \frac{3GM}{2R} \end{aligned}$$

$$\begin{aligned} W &= \int_R^0 \frac{GMmr'}{R^3} dr' \\ &= \left[\frac{GMmr'^2}{2R^3} \right]_R^0 \\ &= -\frac{GMm}{2R} \end{aligned}$$

Hence, work done by gravitational force is $\frac{GMm}{2R}$.